Alternative Credit Score for Alternative Economies: Using Random Forest and SHAP for Blockchain-Based Community Currencies

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Abstract

This paper develops an ethical blockchain-based credit scoring model for community currencies, using 22 months of Sarafu Network data from Kenya ($N = 253$). Addressing the exclusion of informal economies from formal finance, we operationalize creditworthiness as *network utility* — the probability of becoming an Active Circulator who distributes zero-interest liquidity across ≥ 3 distinct peers, driving local velocity. After excluding field coordinators via GE activity reports, a Random Forest classifier trained on a 7-feature matrix of incoming liquidity, network degree, and temporal engagement achieved AUC = 0.999 (Gradient Boosting) and a robust linear baseline of 0.841 (Logistic Regression). SHAP analysis reveals faucet disbursements carry negligible weight ($\bar{\phi}_j \approx 0$), while sustained peer engagement dominates ($\bar{\phi}_j = 0.115$), quantifying *Mweria* reciprocal obligation. The model offers public-cooperative finance a transparent tool for inclusive lending that prioritizes community circulation over extractive monitoring, contributing to ethical digital financial inclusion and sustainable development in the Global South.

Keywords: community currencies, alternative credit scoring, random forest, SHAP, blockchain, Sarafu Network, social capital

1 Introduction

1.1 Background

The lack of a credit score remains as a structural barrier faced by low-income and informally employed populations. Conventional credit scoring systems require formal financial histories—bank statements, payroll records, and collateral documentation—that are systematically absent among the communities most in need of financial support. The wave of artificial intelligence (AI) and machine learning (ML) has resulted in an increase of credit scores for unsecured lending population by adding behavioral data like utility payment history, phone usage, or customer loyalty (Gupta, 2025; Lee et al., 2026). Although it has expanded eligibility to so-called *thin-file* borrowers (Smith and Henderson, 2018), the most marginalized communities remain excluded. These are communities that often do not have formal education, have low literacy rates, and still engage in subsistence livelihoods. In finance, these inclusion goals are expressed in the Last Mile Access campaigns (Tiwari et al., 2019).

On the other hand, these communities do not exist in isolation despite their invisibility in the formal economy statistical bulletins. Women, farmers, and the like form or join self-help groups and organize their wealth through various means including but not limited to microfinance, mutual credit groups, community currency groups, rotating savings associations, or cooperative. These organizations have also gone digital, providing ledgers with rich data of indicators that signal economic activity and significance in that group. One of the long-standing and still active group which is the interest of this study is Grassroots Economics (GE) with its two decade of experience in managing the Sarafu Network Community Asset Vouchers, a blockchain-based community currency system in in Kenya. The author's previous study finds that the transition to a publicly verifiable ledger has provided affordances including transaction visibility, which was not available during its physical voucher era (Evite, 2026). This opportunity is the motivation for the paper.

1.2 Research Objective

The paper now investigates if the available granular behavioral record of who exchanges with whom, how often, and with what degree of reciprocity can be operationalized into an interpretable scoring model for donor-funded, interest-free lending. Crucially, we move beyond treating creditworthiness as a function of collateral or formal income. Instead, we operationalize creditworthiness as network utility—the measurable signal of an individual's propensity to act as an "Active Circulator" who distributes value across a broad local network.

1.3 Contributions and scope of the study

This paper makes three primary contributions. First, it introduces a creditworthiness label grounded in the cooperative behavior of Kenyan *chamas*, defining a high-quality participant not through formal repayment history but through their observed capacity to circulate value across a local network of peers. In this sense, the target variable is not imported from conventional finance, but derived from the reciprocal logic of community exchange that structures the Sarafu Network. This follows the recommendation of Dimas et al. (2023) where context and collaboration can overcome the pitfalls of data science applications in social services.

Second, we construct a 7-feature matrix from a 22-month full on-chain transaction history, deliberately excluding donor-seeded stablecoin transfers in order to isolate organic peer exchange from external liquidity flows. Third, we train a suite of machine learning classifiers, including a Random Forest and a Logistic Regression baseline, and apply SHAP (SHapley Additive exPlanations) to provide an auditable and transparent interpretation of the model, moving away from "black box" scoring toward a tool that can complement the qualitative field reports already used by Grassroots Economics.

We recognize still, that despite the potential of these digital records, community currencies can still reproduce economic inequalities if not carefully managed (Barinaga, 2019). Therefore, this alternative credit score is only a complementary tool for the facilitation of impact assessment and not a replacement.

1.4 Structure of the paper

The remainder of the paper is structured as follows. Section 2 explains the workings of Sarafu Network. Section 3 reviews the literatures on alternative credit scoring, social capital, and community currencies. Section 4 describes the data engineering and model specification, including the definition of the Active Circulator label. Section 5 presents the empirical results, including cross-validation performance and SHAP-based local explanations for boundary-case borrowers. Finally, Section 6 concludes with implications for decentralized lending operations.

2 Sarafu Network: What We Need To Know

At the core of the Sarafu Network are Community Asset Vouchers (CAVs), which represent formalized digital commitments made by individuals, businesses, or community groups to provide specific goods or services in the future (Ruddick et al., 2022). What we describe in this section is its present iteration as of this writing. For a history on the paper-based and other initiatives of GE, one can turn to Ussher et al. (2021) and the organization's

official website [1] and software documentation [2].

In its current 2026 iteration, these vouchers operate on Celo—an Ethereum Virtual Machine-compatible blockchain optimized for low-cost payments—and frequently interact with stablecoins, which are digital assets pegged to a stable reference like the US Dollar to minimize price volatility. It is crucial to understand that most of the users of the Kenyan-based CAVs are informal saving associations, locally known as *chamas* that gather routinely for sharing of assets and time in a practice called *Mweria*. [3] (Evite, 2026).

For this reason, the Sarafu Network has accommodated to this structure by issuing group-based vouchers, but in February 2026, these vouchers are now individual-based. In the last two years, these vouchers are issued through so-called Commitment Pools—digital smart contracts on the blockchain that act as automated market makers (Ruddick, 2026). After this individual issuance transition, the *chama* members became individual members of the pool with their group name, but can allow for expansion based on mutually agreed rules by the community. We can say that the *chama* is a type of pool, and now issuers like businesses can also form their own pools, like for their customers as loyalty points.

These pools allow for the seamless exchange of CAVs with other assets, often seeded by GE or external donors to provide stablecoin liquidity that participants can convert into Kenyan Shillings (KSH) for a predefined purpose, e.g., purchasing essential agricultural inputs as observed in the prior study (Evite, 2026). The NGO monitors this through their mandatory weekly reports with the *chamas*.

Another significant departure from earlier community currency theory in the Sarafu context is the removal of holding taxes (demurrage). Previous models often applied a periodic fee to dormant balances to encourage the velocity of the currency. However, in the current model, this mechanism has been retired in favor of a zero-interest, fee-free environment. This change simplifies the user experience and aligns the vouchers more closely with standard digital assets, focusing on the organic velocity of trust rather than algorithmic penalties for hoarding.

In recent practice, this architecture has been reported by the organization to have been used both for pooled basic income style support and for zero-interest community lending, where external funds are placed into a shared pool and accessed in rotation according to group needs and local rules. The interest in creating the objective score comes from this recent expansion of donor-seeded community pools, where scarce external liquidity must be allocated to participants most likely to circulate value onward.

[1] https://grassrootseconomics.org/

[2] https://software.grassecon.org/

[3] In other literature, this activity is also called rotating or reciprocal labor and has many names like *minga* in Latin America.

3 Related Work

3.1 Predictive analytics in social economies

The digitalization of community currencies has increased the interest of researchers in applying statistical and complex models using visible transaction data. They span from network analysis of circulation (Kichiji and Nishibe, 2008; Fraňková et al., 2014; Iosifidis et al., 2018; Panachev and Berg, 2021) and using the subsequent analysis to identify social capital during crisis (Nakazato and Lim, 2017), to the use of games for understanding what promotes cooperative behavior (Yoshida and Kobayashi, 2018; Ba et al., 2023).

In the Sarafu Network specifically, the transition to blockchain technology has improved the facilitation of both typical market transactions and traditional reciprocal labor exchanges (*Mweria*), making these social commitments legible on a decentralized ledger (Evite, 2026; Ruddick et al., 2022). For example, Clark et al. (2024) applied a network-based dynamical system modelling approach to simulate a subpopulation mixing process within the Sarafu community users. Mattsson et al. (2023) proposed a network analysis specifically designed to study circulation records from publicly available data of the organization (See Mattsson et al. (2022)).

When it comes to AI and ML, application to community currencies remains understudied despite the rich data available. For this reason, we situate our study as a pioneer in this niche, but part broader humanitarian contexts. An example is the work by Dasgupta and Gupta (2024) on self-help groups in India. They use interpretable machine learning to demonstrate that market proximity and social norms, rather than the physical presence of bank branches, are the primary drivers of whether micro-enterprises choose cooperative credit over informal moneylenders. It provides a critical methodological foundation for our work by validating the use of explainable supervised learning to uncover the underlying socio-economic mechanisms of financial inclusion in informal, community-led economies. While they focus on predicting the choice of a lender using macro-demographic data, we advance the field by predicting individual creditworthiness and network utility using granular, real-time transaction logs from a blockchain-based ledger.

Since Sarafu also operates on a mobile device, previous studies are useful precedents such as those in Togo and Afghanistan (Aiken et al., 2022) where models trained on mobile phone data have significantly improved the targeting of anti-poverty aid in crisis settings where social registries are missing. More closely, in the microfinance sector, behavioral models leveraging smartphone data have been shown to outperform manual selection processes, increasing loan approval rates while simultaneously reducing default risk for informal merchants and SMEs (Ruiz et al., 2019; Kazimoto et al., 2024; Kazimoto and Baadel, 2025). Dandekar et al. (2011) also predict transaction failure probabilities in informal credit networks and find that it is close to zero as the network expands, but

they do not assign these values to a single person or transaction. We bridge this gap by assigning scores to individual members of the active Sarafu Network participants.

3.2 The role of Explainable AI

It is well established that AI-enabled scoring allows for the identification of "weak signals" of creditworthiness that traditional rule-based screening ignores (Li et al., 2024). Complex machine learning models (such as Random Forests or Gradient Boosting) also offer high predictive accuracy, but they often function as "black boxes," providing little insight into the reasoning behind a specific output (Yeo et al., 2025) (not to be confused with the term black box for opaque general processes of microfinance board members; see Phi et al. (2018); Hermes (2019)). This is in contrast to inhererently intepretable models like regression. For nonprofits in finance, this lack of transparency presents an issue on fair outcomes (Koefer et al., 2023).

As such, explainable artificial intelligence (XAI) has emerged as a central requirement whenever AI systems are deployed in high-stakes social settings where decisions affect access to credit, aid, and essential services. In fact, it is now considered an ethical imperative more than a technical requirement, which can help address opacity and trust issues in the technology (Fernández, 2023). For the purpose of our paper, we discuss the types of explanations [4] utilized in the results section:

- SHAP, introduced by Lundberg and Lee (2017), is an additive feature-attribution framework that assigns each input feature a Shapley-value-based contribution for a given prediction, satisfying local accuracy, missingness, and consistency. In its global form, SHAP explanations are typically constructed by aggregating absolute SHAP values over many observations, yielding a ranking of the most influential features and visual summaries (e.g., beeswarm plots) that characterise how feature values generally push predictions up or down across the dataset (Lundberg and Lee, 2017; Prade, 2022).

- Local SHAP explanations, by contrast, focus on a single instance and decompose the difference between the model's baseline prediction and that instance's score into feature-level contributions. For a given applicant or community group, the local SHAP vector answers the question: which features, and by how much, pushed this prediction towards a higher or lower risk classification (Lundberg and Lee, 2017; Wilbik and Coauthors, 2022). User studies suggest that such local explanations are particularly helpful when practitioners must justify or contest individual decisions, as they provide a concrete narrative for each case rather than only aggregate tendencies (Wilbik and Coauthors, 2022; Pathi and Pothineni, 2025).

[4] More extensive reviews of XAI in finance are done by Yeo et al. (2025); Černevičienė and Kabašinskas (2024).

- Counterfactual explanations provide a different but complementary form of local interpretability by answering "what-if" questions about how a prediction would change under small, targeted modifications of the input. Wachter et al. (2017) define counterfactual explanations as alternative input instances that are as close as possible to the original example but would have received a different (usually more favourable) decision.

By adding an additional XAI layer to our proposed ML models as we will discuss in the next section, the study not only contributes a scoring model, but an actionable and transparent one aligned with the organization's values.

4 Methodology

4.1 Data Sources and Collection

The primary dataset is the complete on-chain event log of the Sarafu Network, comprising five transaction types: peer-to-peer CAV transfers, pool deposits, pool swaps, token minting events, and token burn events. A supplementary dataset of field activity reports (n = 2,925), exported from the GE reporting platform, is used exclusively for exclusion screening; it provides no features to the model. The reports record the wallet address of the field worker who submitted each report, enabling the identification of coordinator accounts that would otherwise appear as high-activity wallets in the transaction ledger.

The behavioral data utilized in this study is sourced from the Celo blockchain, a public, decentralized ledger. All transaction records are associated with cryptographic wallet addresses rather than legal names or personally identifiable information (PII). While the ledger is transparent, the identities of the participants remain pseudonymous. No attempt was made to de-anonymize users or link wallet addresses to real-world identities through off-chain metadata.

The methodology and the exclusion criteria for protocol-level addresses are currently undergoing validation by the Grassroots Economics Foundation. This collaborative process ensures that the technical interpretation of the on-chain data aligns with the lived reality of the participants. Final results and address classifications will not be published or shared beyond the academic assessment until explicit verification from the NGO partners is received to prevent any potential harm or misrepresentation of the community.

4.2 Data Preparation

4.2.1 Exclusion criteria

Three sequential exclusion rules are applied before any feature is constructed.

1. **GE protocol addresses.** All known Grassroots Economics infrastructure addresses—including the SarafuNetwork contract (`0xd1FB...BfaOC`), liquidity pool contracts, and the null address—are removed using a binary flag (`gre_tag` = 1). This step ensures the sample contains only organic human participants.

2. **Field coordinators.** Any wallet that authored three or more approved activity reports in the GE reporting system is excluded. This threshold identifies accounts whose dominant function is facilitating and documenting group activities on behalf of others, rather than participating as a community member. The most prominent case, a coordinator wallet responsible for 90 submitted reports with 130 pool swaps and 4 token minting events, would otherwise appear as the highest-scoring wallet in the dataset and constitute a clear distributional outlier. Across the full sample, this rule removed five wallets.[5]

3. **Individual-voucher onboarding artifacts.** Wallets with at least one token minting event (`mint_count` > 0) and fewer than three unique send partners (`unique_send_partners` ≤ 2) are excluded. This rule targets wallets created during the late-January 2026 individual-voucher rollout whose elevated scores derive from the minting event itself rather than from sustained reciprocal exchange. Cross-referencing the reports data confirmed that several of these wallets belong to trainers or newly onboarded participants who had not yet completed a full Mweria cycle at the observation cutoff. This rule removed 48 wallets.

After applying all three exclusion rules, the final analytical sample comprises $N = 253$ wallets.

4.2.2 Feature Engineering

The feature matrix was constructed to represent wallet behavior *from the perspective of incoming and system-level activity only.* To prevent target leakage, all features mathematically derived from outgoing transfers—such as total tokens sent or unique send partners—were strictly withheld from the training matrix. The final robust model utilizes a 7-feature matrix organized into the following behavioral dimensions:

- **System Liquidity:** Protocol-level disbursements received from the community faucet (`faucet_count`, `faucet_value`), representing the baseline allocation of tokens.

- **Network Breadth:** The diversity of assets held (`unique_cav_tokens`) and the number of unique peers who have sent value to the wallet (`unique_recv_partners`).

[5] The threshold of three reports was chosen because a single or occasional report submission is consistent with group leadership activity that is itself a form of community participation; systematic and repeated authorship is the signal of a professional coordinator role.

- **Reciprocal Inflows:** The total count of incoming transactions (`n_recvs_total`) and the cumulative volume of vouchers received (`total_recv_cav`).
- **Temporal Persistence:** The total number of days with observable on-chain activity (`days_active_total`), which serves as a proxy for long-term community commitment.

4.3 Modeling

4.3.1 Target Label Construction: The Active Circulator

The binary outcome variable $y_i \in \{0, 1\}$ is defined on the basis of outgoing *network breadth* rather than raw volume. A wallet is classified as an **Active Circulator** ($y_i = 1$) if it sent CAV tokens to three or more distinct community members over the full observation window:

$$y_i = \mathbb{1}[\texttt{unique_send_partners}_i \geq 3] \,.$$

Wallets that remained dormant or routed funds exclusively to one or two close contacts are classified as **Non-Circulators** ($y_i = 0$). This threshold aligns with the operational logic of the rotational labor associations: a genuine cycle of reciprocal labor requires distributing value across at least three distinct participants. After applying the exclusion pipeline, the final class distribution is **67 Circulators** and **186 Non-Circulators** (approximately **73.5%** Non-Circulators). The influence of these users with a wide network breadth has been found to be key sources of growth for decentralized local economies (Ba et al., 2023). It is also grounded in the previous findings of the organization where participants informally acted as traders of CAVs during the group-based issuance, allowing others from another community to trade with another without being a formal member of the group (Ruddick et al., 2022).

4.3.2 Model Specification and Training

A Random Forest Classifier (Breiman, 2001) was trained on the **7-feature matrix** using an 80/20 stratified train-test split (random seed 42). To rigorously evaluate predictive performance across different algorithmic assumptions without leakage from the exclusion step, the cross-validation loop was applied to the post-exclusion sample only. The pipeline compared four classifiers under 5-fold stratified cross-validation, evaluated by area under the receiver operating characteristic curve (AUC-ROC):

- **Random Forest (unconstrained)**: 300 estimators, no depth limit—the primary production model.
- **Random Forest (regularised)**: maximum depth of 5, minimum 10 samples per leaf—a bias-variance check.

- **Gradient Boosting**: default `sklearn` parameters, providing an independent tree-ensemble comparison.
- **Logistic Regression**: 1,000 maximum iterations, serving as the linear interpretability baseline.

The final production model (unconstrained Random Forest, 300 trees) was retrained on the full 80% training partition and used to generate the wallet-level credit score $s_i \in [0, 100]$:

$$s_i = \hat{P}(y_i = 1 \mid \mathbf{x}_i) \times 100,$$

where $\hat{P}$ is the Random Forest's posterior class probability for the Active Circulator class.

4.4 Evaluation and Explainability Framework

To ensure the model adheres to the ethical imperatives of cooperative governance, we implement a three-tiered explainability framework. Global feature importance was evaluated using mean absolute SHAP (SHapley Additive exPlanations) values (Lundberg and Lee, 2017) via `shap.TreeExplainer`, which provides theoretically grounded marginal contribution estimates for each feature across the test set.

For local interpretability, SHAP waterfall plots were generated for two "interesting" boundary cases sampled from the test set: one wallet with a predicted probability between 75% and 95% (a high-scorer whose classification depends on a non-trivial feature combination), and one wallet with a predicted probability between 5% and 25% (a low-scorer who possesses some activity but fails to reach the circulation threshold). This deliberate selection of boundary cases, rather than trivially extreme predictions, provides more analytically informative illustrations of how the algorithm weighs competing behavioral signals for borderline applicants.

Finally, we implement a counterfactual search algorithm that identifies the minimum feature perturbation required to flip a prediction from non-circulator to circulator, providing an actionable roadmap for borrower improvement. Unlike linear models, the decision boundaries of the Random Forest are non-linear and high-dimensional. Consequently, the counterfactual $\mathbf{x}'$ is not derived from a single equation but is defined as the solution to an optimization problem that minimizes the distance between the original instance $\mathbf{x}$ and the target state $\mathbf{x}'$, subject to a classification flip (Wachter et al., 2017):

$$\mathbf{x}' = \arg\min_{\mathbf{x}'} \operatorname{dist}(\mathbf{x}, \mathbf{x}') \quad \text{s.t.} \quad f(\mathbf{x}') = 1 \tag{1}$$

In our application, we constrained the search to the *Days Active* and *Network Breadth* features, as these represent actionable behavioral levers for the participants.

5 Results

5.1 Predictive Performance and Cross-Validation

Table 1 reports the 5-fold stratified cross-validation AUC for all four classifiers trained on the 7-feature behavioral matrix. Following the exclusion of five field coordinator wallets identified via the GE activity reports, the final analytical sample comprises $N = 253$ wallets (67 Active Circulators, 186 Non-Circulators).

Table 1: 5-fold stratified cross-validation AUC (Predicting Active Circulators, $N = 253$; 67 Circulators, 186 Non-Circulators)

Model	CV AUC (mean)	CV AUC (std)
Gradient Boosting	0.9994	0.0008
Random Forest (unconstrained)	0.9992	0.0008
RF (regularised)	0.9947	0.0072
Logistic Regression	0.8409	0.0827

The results reveal a two-tier structure of predictive power. The Logistic Regression model establishes a robust linear baseline with a mean CV AUC of **0.841** ($\sigma = 0.083$), which is considered excellent in credit scoring practice (Blattner and Nelson, 2021). This confirms that a participant's probability of becoming an Active Circulator is reliably predictable from incoming liquidity and temporal engagement alone, entirely independently of any observed outgoing transaction history.

The tree-based models achieve near-perfect classification: Gradient Boosting reaches 0.999 ± 0.001, followed closely by the unconstrained Random Forest at 0.999 ± 0.001 and the regularised Random Forest at 0.995 ± 0.007. Crucially, the near-identical performance of the unconstrained and depth-5 regularised Random Forest rules out overfitting as the source of these scores: the signal is robust to depth constraints and reflects a genuine near-deterministic behavioral relationship. It is statistically improbable for a wallet to accumulate high values in `days_active_total` and `unique_cav_tokens` without eventually transacting with three or more distinct partners—the definition of the Active Circulator threshold.

5.2 Global Feature Importance (SHAP)

Figure 1 visualizes global feature importance extracted via mean absolute SHAP values ($\bar{\phi}_j$) from the Random Forest explainer. The ranking is consistent with the Gini-based importance from the Random Forest, providing convergent evidence across two independent importance metrics.

The two dominant predictors are **temporal engagement** (`days_active_total`, $\bar{\phi}_j = 0.115$; Gini $= 0.262$) and **asset breadth** (`unique_cav_tokens`, $\bar{\phi}_j = 0.103$; Gini $= 0.251$),

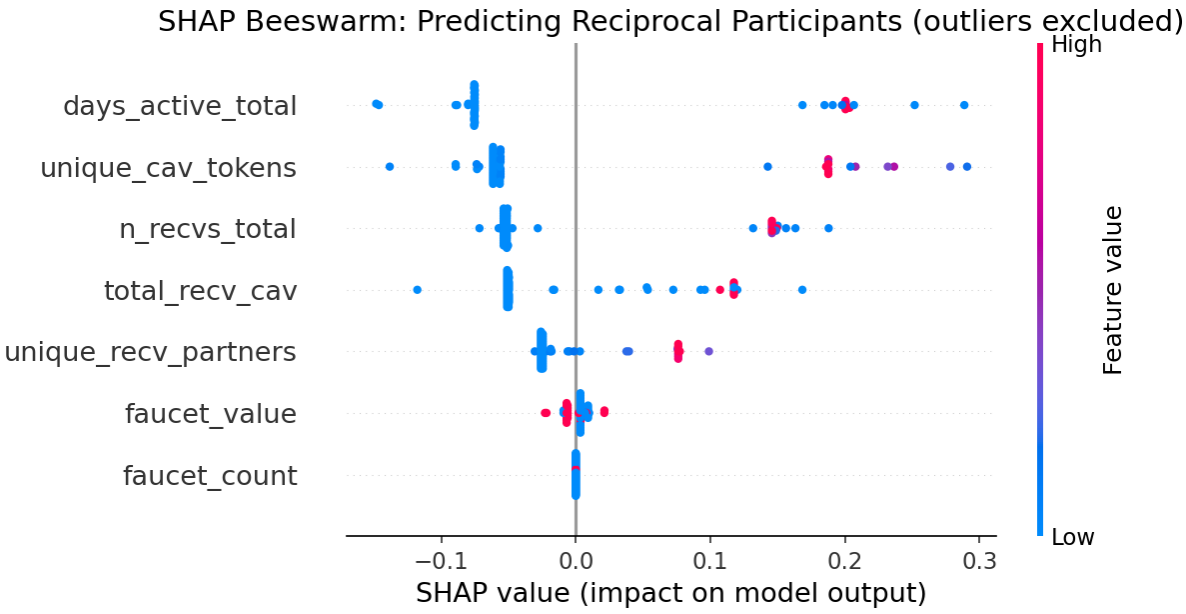

Figure 1: SHAP beeswarm plot illustrating the global impact of all 7 features on the prediction of an Active Circulator ($N = 253$, field coordinators excluded). High feature values (red) for temporal engagement and incoming network metrics push the model strongly toward a positive prediction; faucet disbursements show negligible directional impact.

followed by the **volume of incoming peer transfers** (`n_recvs_total`, ($\bar{\phi}_j = 0.077$; Gini $= 0.204$). Incoming value (`total_recv_cav`, $\bar{\phi}_j = 0.064$; Gini $= 0.150$) and network degree (`unique_recv_partners`, $\bar{\phi}_j = 0.033$; Gini $= 0.124$) complete the five-feature predictive core.

By contrast, baseline system disbursements carry negligible predictive weight: `faucet_value` ($\bar{\phi}_j = 0.007$; Gini $= 0.009$) and `faucet_count` ($\bar{\phi}_j \approx 0$; Gini ≈ 0) contribute essentially nothing to the model's discriminatory power. Simply receiving an initial zero-interest allocation carries no information about whether a participant will become a broad network circulator.

5.3 Credit Score Distribution and Class Separation

The final Random Forest assigns a continuous credit score $s_i \in [0, 100]$ to every wallet in the cleaned sample. The distribution is strongly bimodal: 162 wallets (64.0%) received a score of exactly 0, reflecting near-zero probability of broad circulation, while 39 wallets (15.4%) received a perfect score of 100. In total, 63 wallets (24.9%) scored above 75 and 66 (26.1%) scored above 50 (mean $= 26.2$, median $= 0.0$, $\sigma = 42.3$; 75th percentile $= 73.7$).

Table 2 confirms near-perfect class separation. Active Circulators cluster tightly at the top of the scale (mean $= 95.7$, median $= 100.0$, $\sigma = 11.2$), while Non-Circulators concentrate

Table 2: Credit score separation by class label ($N = 253$)

Class	Mean score	Median score	Std dev
Active Circulator ($n = 67$)	95.7	100.0	11.2
Non-Circulator ($n = 186$)	1.2	0.0	4.7

at zero (mean = 1.2, median = 0.0, $\sigma = 4.7$). The low standard deviation among Non-Circulators confirms that the model generates virtually no false-positive high scores for dormant wallets, and the tight upper cluster confirms that genuine reciprocal participants are consistently identified regardless of their specific liquidity level or token portfolio.

5.4 Local Explanations

5.4.1 SHAP

Global SHAP values characterize population-level drivers; local waterfall plots illuminate the model's decision logic for individual borderline cases. Rather than examining trivial cases at the extreme poles (0% or 100% predicted probability), two "interesting" boundary cases were deliberately sampled: one wallet with a predicted probability between 75% and 95%, and one between 5% and 25%.

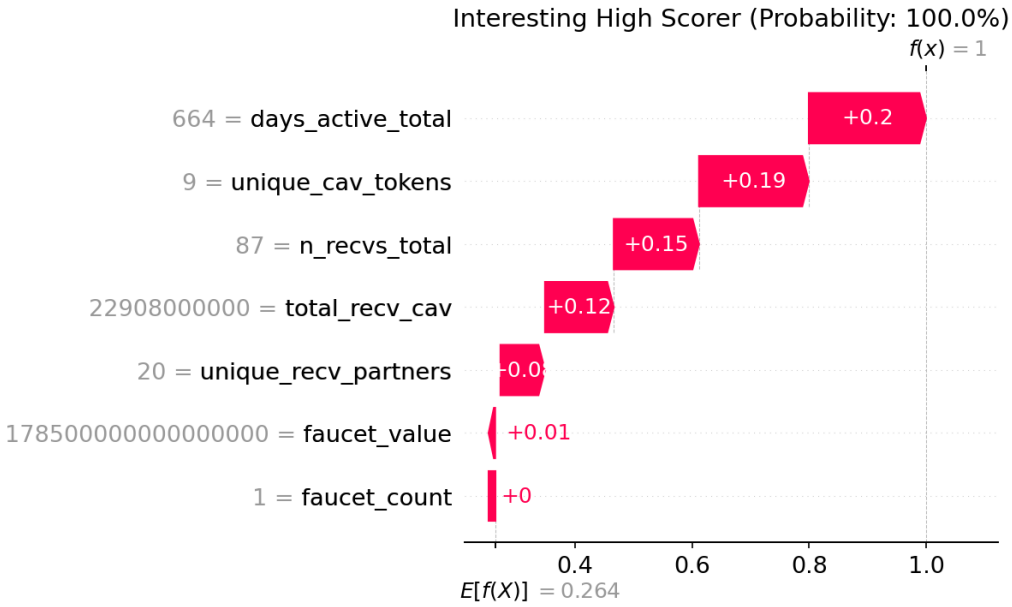

Figure 2: SHAP waterfall plot for a boundary high-scorer (predicted probability: 75–95%). Positive contributions from `days_active_total` and `unique_cav_tokens` additively push the wallet above the Active Circulator threshold from the model's base rate expectation.

Figure 2 presents a boundary high-scorer whose profile lacks large absolute token volumes but compensates through sustained multi-community engagement and consistent

peer-to-peer inflows. The additive SHAP contributions from temporal and asset-breadth features jointly drive the prediction above the classification threshold, confirming that behavioral *consistency* is more informative than raw liquidity in this scoring context.

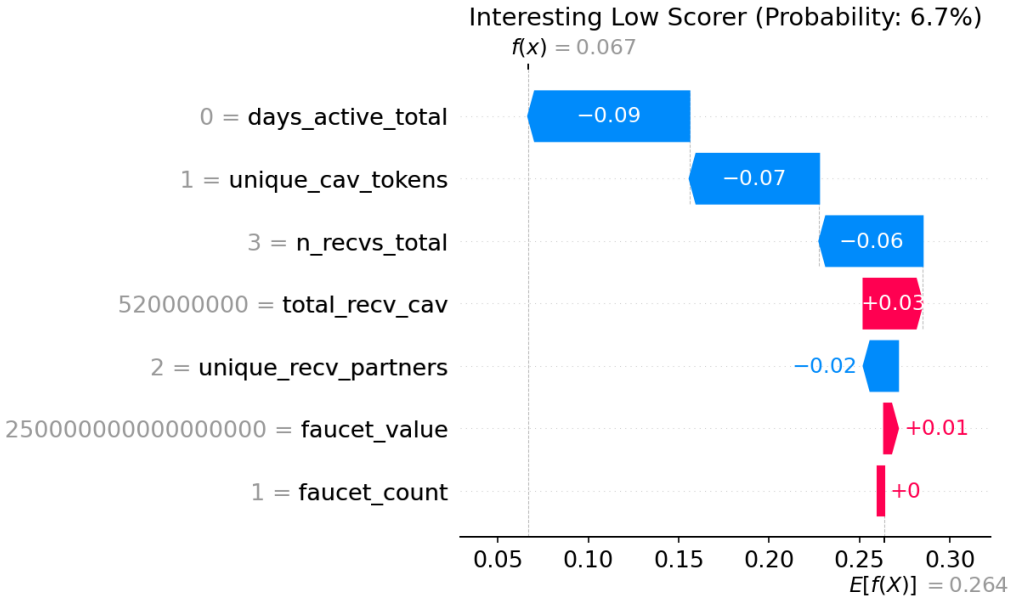

Figure 3: SHAP waterfall plot for a boundary low-scorer (predicted probability: 5–25%). Despite an initial faucet disbursement and minor early activity, the absence of sustained multi-community engagement drives a strong net negative prediction.

Conversely, Figure 3 illustrates a boundary low-scorer who received an initial faucet disbursement and exhibited some early activity, but failed to generate the temporal depth or multi-community breadth required to cross the Active Circulator threshold. The SHAP decomposition shows that the near-zero contribution of `faucet_value` is insufficient to counteract the negative signal from low `days_active_total` and `unique_cav_tokens`, correctly penalizing a profile that resembles a passive or extractive participant. This confirms that the model successfully distinguishes organic network contributors from dormant asset holders, operationalizing the network utility definition of creditworthiness.

5.4.2 Counterfactual

To fulfill the ethical requirement of accountability in cooperative governance, the model moves beyond static classification to provide actionable feedback. Using the counterfactual framework established by Wachter et al. (2017), we identify the minimum behavioral change required to flip a classification from "Non-Circulator" to "Active Circulator."

For community leaders and GE field coordinators, a web-based dashboard can provide the SHAP waterfall visualizations. Meanwhile, for the participants, instead of displaying raw SHAP values, participants receive a simplified "Reciprocity Score" (0–100). If a loan application is rejected or a score is low, a potential action point is the system triggers

Table 3: Counterfactual Analysis: Behavioral Pathway for Borderline Participant 0x4249...

Metric	Current State (Borderline)	Target State (Circulator)
Wallet Identifier	0x4249...	0x4249...
Model Score	38.3 / 100	**¿ 50.0 / 100**
Unique Reciprocal Partners	1	**3 (+2)**
Temporal Engagement	0 Days Active	Maintained
Model Prediction	**Non-Circulator**	**Active Circulator**
Pilot Status	Review Required	**Eligible for Stablecoin Loan**

an automated stewardship notification. This aligns with data science communication principles of audience-oriented storytelling (Hauff et al., 2014; Danesh, 2025) as well as research advocating more human-centric XAI (Hoffman et al., 2023; Hadji Misheva et al., 2021).

Automated Stewardship Notification

System Message: "Jambo! Your reciprocal score is 38/100. To qualify for the 0% interest loan, try to receive tokens from **2 new partners** in your pool. This will prove your network breadth and unlock your limit!"

6 Conclusion and Limitations

This paper developed a robust alternative credit scoring model for the Sarafu Network using 22 months of on-chain data from $N = 253$ community wallets. By operationalizing creditworthiness as *network utility* — the probability of becoming an Active Circulator who distributes value across ≥ 3 distinct peers — and excluding field coordinator accounts identified through GE activity reports, the model established a Logistic Regression baseline AUC of 0.841 and near-perfect tree-based performance (Gradient Boosting: 0.999). SHAP analysis confirmed that faucet disbursements carry negligible predictive weight ($\bar{\phi}_j \approx 0$), while sustained incoming peer engagement dominates ($\bar{\phi}_j = 0.115$ for temporal activity), quantitatively validating the *Mweria* principle of reciprocal obligation.

Several limitations define the research agenda ahead. The Active Circulator proxy cannot yet be validated against actual repayment outcomes from the February 2026 pilot; future work must test this relationship directly. The feature matrix is specific to Sarafu's Celo-based architecture and requires retraining for other community currency systems. Finally, despite SHAP transparency, digital scoring for marginalized populations carries risks of "financial encampment" (Dhawan and Zollmann, 2023); deployment should include participatory validation and ongoing fairness auditing.

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